

MANAGEMENT OF ANIMAL BITES

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*Snake bite management is now a
stand-alone guideline.*

1. GENERAL INFORMATION

	HUMAN	CAT	DOG
Injuries	- Intercanine bite > 3 cm suggestive of adult bite – consider NAI. <u>Clenched fist injuries</u> against teeth.	<u>Puncture</u> wounds – inoculate bacteria into deep tissue with innocuous looking wounds	- Superficial to deep tissue injuries <u>Crush</u> injuries – dermis intact but underlying tissue devitalized due to damaged blood supply.
Infection Risk	Highest risk	Moderate risk	Lowest risk

2. CLINICAL ASSESSMENT

- Time of injury (↑ risk of infection for facial wounds >24 hours & non-facial wounds >12 hours)
- Any first aid provided (e.g., irrigation/cleansing of wound)
- Vaccination status (e.g., tetanus, rabies)
- High risk sites of injury
 - Hands/feet – deep tissues that are more superficial; bacteria easily inoculated into deep structures resulting in complicated infection.
 - Skin over cartilaginous structures – poor blood supply with ↑ risk of wound infection & poor healing
 - Genitalia – important organs; area of dependence, prone to edema; proximity to commensals from GI tract
- Extent of injury (surrounding structures: joint, bone, neurovascular structures) & foreign body
- Signs & symptoms of wound infection
 - Erythema, discharge, pain
 - Local spread: lymphangitis, cellulitis, abscess, osteomyelitis, thrombophlebitis, pyomyositis
 - Systemic: Fever + viral-like symptoms

3. SUBSPECIALTY SURGICAL CONSULTATION

- Purpose of surgical consult:
 - Assess need for surgical debridement & exploration of wound.
 - Repair of damaged surrounding structures

BODY PART	SURGICAL TEAM
Face	Plastic surgery
Head & neck	Plastic surgery or ENT
Hand (distal to wrist joint)	HRM
Upper limb or Lower limb	Orthopedics
Torso or Groin	Pediatric surgery Orthopedics (spine)

- Apart from wounds limited to the epidermis, all bites should be consulted with the surgeon

4. INVESTIGATIONS

4.1 IMAGING

IMAGING MODALITY	INDICATION
X-ray (at least 2 views in different planes)	▫ Suspected underlying fracture. ▫ Suspected bone penetration or joint involvement. ▫ Suspected FB within wound (e.g., tooth fragment). ▫ Clenched fist injuries, bites to the hand.
Ultrasound	▫ Evaluation of radiolucent FB. ▫ Deep seated infections/collections.
CT Brain	▫ For suspected penetrating scalp injuries.

4.2 LABORATORY

INVESTIGATION	INDICATION
Wound aerobic & anaerobic c/s Gram-stain & smear	▫ Only for clinically infected wounds ▫ Random wound cultures have limited value in determining infection risk
Tissue aerobic & anaerobic c/s	▫ Debrided tissue of surgically managed wounds
Blood aerobic c/s Inflammatory markers (FBC, CRP)	▫ Patients with systemic features of infection ▫ Immunocompromised patients

5. MANAGEMENT OF MAMMALIAN BITES

5.1 WOUND CARE

5.1.1 **WOUND IRRIGATION (1)**

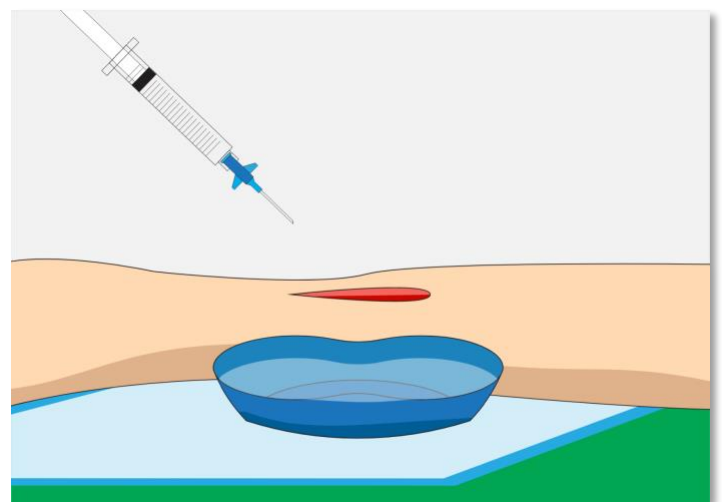
- Proper wound irrigation ↓ bacterial load of bite injuries & removes loose material.
- Inappropriate irrigation pressures could result in wound infection & poor healing.
 - High irrigation pressures = Push surface pathogens deeper into the tissue
 - Low irrigation pressures = Does not overcome the adhesive properties of bacteria
- ↑ Irrigation pressures generated with:
 - ↓ Syringe volume
 - ↑ Cannula gauge
 - ↑ Physical pressure applied.

5.1.1.1 Equipment List

- PPE (gown, gloves, eye shield)
- 20ml syringe
- 22G cannula
- Kidney dish
- Incopad
- Irrigation solution (50ml/cm of laceration or, 50ml/cm³ of wound)
 - Sterile saline
 - Povidone-Iodine: Normal saline (1 part to 9 parts) (contaminated wounds)

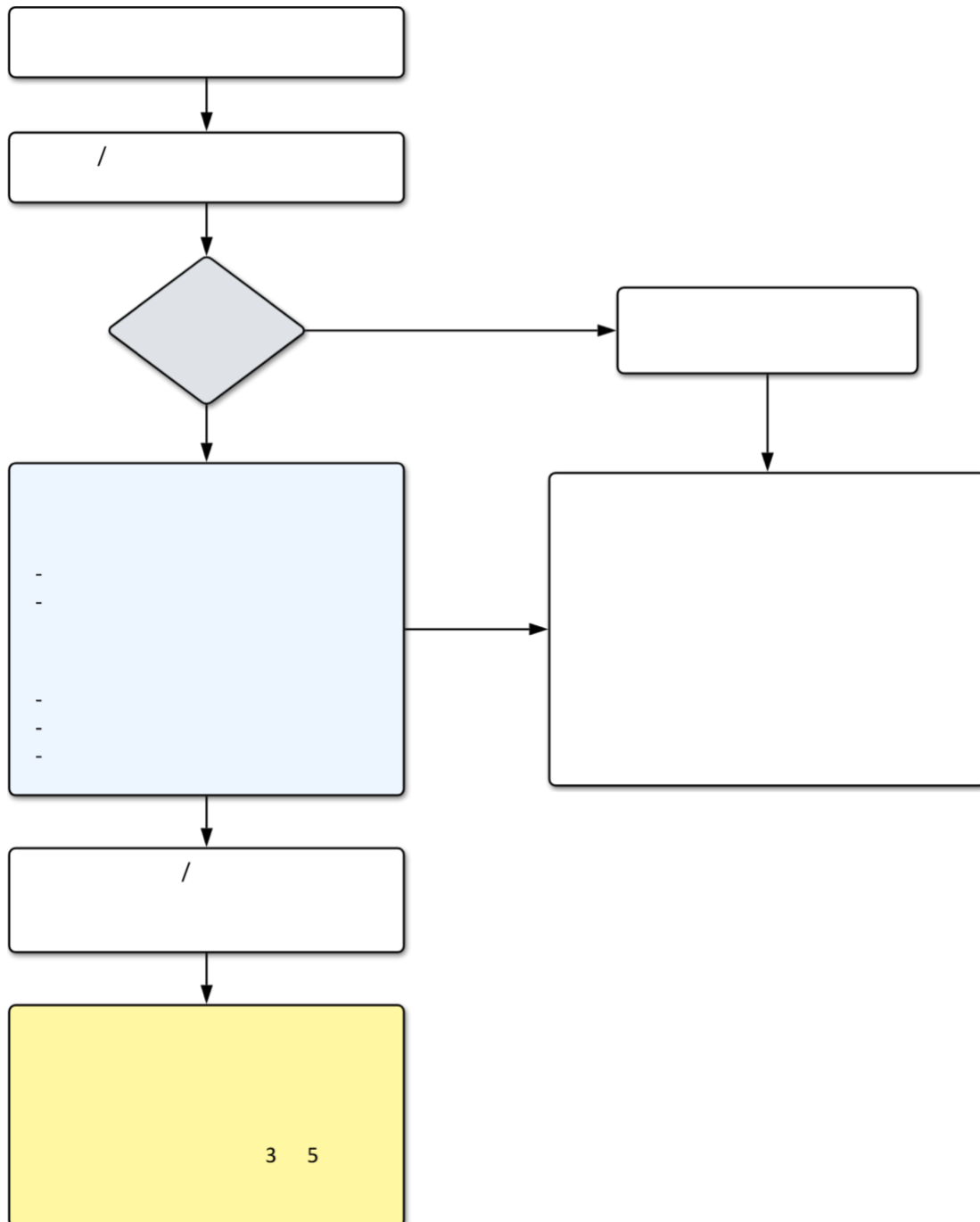
5.1.1.2 How To Irrigate?

- A. Position patient such that fluid flows from cleanest to most contaminated part of the wound
- B. Place kidney dish to catch fluid running off
- C. Hold the filled syringe with the tip of the cannula about 1 inch from the wound.
- D. Instill the solution continuously into the wound, starting from the cleanest area.
- E. Use enough pressure to flush debris out of the wound but avoid splashing/squirting fluid.
- F. Do not force fluid into wound pockets.
- G. Irrigate with sufficient volume.



5.1.2 CLOSING THE WOUND

- **DO NOT** close a bite wound with tissue glue (adhesive seals bacteria in & does not allow collections to drain).
- **DO NOT** close infected wounds.
- Primary wound closure is **NOT RECOMMENDED** for bite wounds.



5.2 ANTIBIOTICS

- Antibiotics must have good activity against *Pasteurella* spp. & anaerobes.
- Drugs with low activity against *Pasteurella multocida* include first-generation cephalosporins, macrolides, penicillinase-resistant penicillin (cloxacillin), aminoglycosides.
- Augmentin is the first line antibiotic of choice (refer to [ANNEX A](#) for alternatives)

5.2.1 ANTIBIOTIC PROPHYLAXIS

- Duration of prophylaxis = 3 days

	Break in skin, draw blood	Break in skin, no blood	No break in skin
Human	Yes	Consider prophylaxis. ▫ High risk area [^] ▫ Significant co-morbidity [#]	No
Dog	Yes, offer prophylaxis. ▫ If there is penetration of bone, joint, vascular structure. ▫ Deep, crush, significant tissue damage. ▫ Highly contaminated (e.g., dirt or tooth in the wound)	No	No
	Consider prophylaxis. ▫ High risk area [^] ▫ Significant co-morbidity [#]		
Cat	Yes	Consider if the wound is suspected to be deep	No

[^]High risk areas: hands, feet, genitals, skin overlying cartilaginous structures, pre-existing or resultant edema.

[#]Significant co-morbidities: Immunocompromised host (asplenia, diabetes, decompensated liver disease).

5.2.2 EMPIRIC ANTIBIOTICS

- Duration of empiric antibiotics = 5 days
- Collect microbiological samples prior to treatment.
 - Swab discharge for wound aerobic & anaerobic cultures.
 - Debrided tissue for tissue aerobic & anaerobic cultures (remind surgeons).
- If discharged from CE, early review for wound inspection in 1 – 2 days.

5.2.3 INTRAVENOUS ANTIBIOTICS

- Systemic signs of infection (fever, abnormal vital signs)
- Complicated infection (septic arthritis, osteomyelitis, tenosynovitis, bacteremia)
- Rapidly progressing erythema
- Progression of signs/symptoms after 48H of oral antibiotic therapy

5.3 POST-EXPOSURE PROPHYLAXIS

5.3.1 TETANUS TOXOID VACCINE

Table from "Red Book®: 2015
Report of the Committee on
Infectious Diseases, 30th Edition"

Hx Of Adsorbed Tetanus Toxoid	TYPE OF WOUND			
	Clean & Minor Wounds		All Other Wounds ^a	
	DTaP, Tdap or Td ^b	TIG ^c	DTaP, Tdap or Td ^b	TIG ^c
< 3 doses or unknown	Yes	No	Yes	Yes
≥ 3 doses	No, if < 10 years since last tetanus- containing vaccine	No	No, if < 5 years since last tetanus- containing vaccine	No
	Yes, if ≥ 10 years since last tetanus- containing vaccine dose	No	Yes, if ≥ 5 years since last tetanus- containing vaccine dose	No

Tdap indicates booster tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis vaccine; DTaP, diphtheria and tetanus toxoids and acellular pertussis vaccine; Td, adult-type diphtheria, and tetanus toxoids vaccine; TIG, Tetanus Immune Globulin (human).

^aSuch as, but not limited to, wounds contaminated with dirt, feces, soil, and saliva; puncture wounds; fracture avulsions; and wounds resulting from missiles, crushing, burns and frostbite.

^bDTaP is used for children < 7 years. Tdap is preferred over Td for under-immunized children ≥ 7 years who have not received Tdap previously.

^cIntravenous immunoglobulin (IVIg) should be used when TIG is not available.

More frequent boosters are not needed and can accentuate adverse effects.

5.3.2 RABIES POST-EXPOSURE PROPHYLAXIS (PEP)

- Though Singapore has been rabies free since 1953, returning travelers with animal bite injuries may require rabies PEP.
- There is no time limit for the administration of rabies PEP.
- Rabies PEP comprises of 3-steps:
 - Local wound management
 - Rabies vaccination
 - Human rabies immunoglobulin (HRIG) if indicated

5.3.2.1 Local Wound Management

- Immediate wound irrigation & cleaning for a minimum of 15 minutes with clean water & soap.
- Irrigate with virucidal agents like povidone-iodine if available.
- If not performed after wound was sustained, wound irrigation & cleaning should be performed (regardless of the duration from injury at presentation)

5.3.2.2 Rabies Vaccination

- Vaccination should be administered in the deltoid, or lateral quadriceps.
- Avoid gluteus muscle

5.3.2.3 Human rabies immunoglobulin (HRIG)

- As per table below.
- It may be administered up to and including D7 of PEP series.
- Refer to [“Rabies Post-Exposure Prophylaxis Protocol”](#)

WHO Post-Exposure Rabies Management

WHO Exposure Category	Local Tx	Immunization Status	PEP Regime	RIG
Category I Touching or feeding of animals; licks on intact skin	Required	NA	Not required	Not required
Category II Nibbling of uncovered skin, minor scratches, or abrasions without bleeding. If bat exposure, treat as Cat III.	Required	No prior immunization received.	Day 0, 3, 7, 14	Not required
		Previously immunized:		
		▫ 4-dose PEP (LD* < 3 months)	Not required	Not required
		▫ All other regimes including 4-dose PEP (LD* > 3 months)	Day 0, 3	Not required
		Immunocompromised (regardless of previous vaccination).	Day 0, 3, 7, 14, 28	Required
Category III Single or multiple transdermal bites or scratches, licks on broken skin; Contamination of mucous membrane with saliva from licks, contacts with bats. Bites to the head, neck, face, hand & genitals are considered Cat III due to extensive innervation.	Required	No prior immunization received.	Day 0, 3, 7, 14	Required
		Previously immunized:		
		▫ 4-dose PEP (LD* < 3 months)	Not required	Not required
		▫ All other regimes including 4-dose PEP (LD* > 3 months)	Day 0, 3	Not required
		Immunocompromised (regardless of previous vaccination).	Day 0, 3, 7, 14, 28	Required

*LD – Last Dose

6. DISPOSITION

- Follow up with surgical subspecialty for wound review (as per consult recommendation).
- Follow up with Pediatric ID outpatient clinic in 1 – 2 days for review (if no surgical subspecialty follow-ups given)

7. HERPES B VIRUS / MONKEY BITE (2) (3)

- B virus (*Cercopithecine Herpes virus 1*) is a zoonotic agent that can cause fatal encephalomyelitis in humans. The virus naturally infects macaque monkeys.
- Mode of transmission: bites, scratches, cage scratches, needlestick injuries
- Evaluate the need for tetanus toxoid booster & rabies PEP.

7.1 HERPES B POST-EXPOSURE PROPHYLAXIS (PEP)

PEP Recommended	PEP Considered	Not Recommended
Broken skin exposure or mucosal surface exposure (with or without injury) to a <u>high-risk source</u> (e.g., macaque monkey that is ill, immunocompromised, or known to be shedding the virus)	Mucosal splash, adequately cleaned	Skin exposure (intact skin)
Inadequately cleansed [^] skin exposure (with loss of skin integrity) or mucosal surface exposure (with/without injury)	Laceration that has been adequately cleansed	Non-macaque species of non-human primates.
Laceration to the head, neck, or torso		
Deep puncture bite		

[^]Adequately cleansed = washed within 5-minutes of injury for a duration of more than 15 minutes.

7.2 ANTI-VIRAL DOSING

Anti-Viral Agent	Prophylaxis (PO)	Treatment (IV)	
	14-day Course	With CNS symptoms	No CNS symptoms
*Valacyclovir	<u>20 mg/kg</u> (max. 1g) 8H	-	-
Acyclovir	<u>20 mg/kg</u> (max. 800mg) 5x/day	<u>10 – 15 mg/kg</u> 8H	-
Ganciclovir	-	<u>5 mg/kg</u> 12H	<u>5 mg/kg</u> 12H

*Valacyclovir is the preferred antiviral for prophylaxis

7.3 TREATMENT OF HERPES B

- There is no need for serological testing / confirmation of Herpes B in asymptomatic patients that are planned for discharge with anti-viral prophylaxis
- Pediatric Infectious Diseases service should be consulted for suspected symptomatic Herpes B infection. Initiation of treatment and investigations should be performed after discussion with Paeds ID.

	SYMPTOMS
EARLY SYMPTOMS	▫ Vesicular or ulcerative lesions, pain or itching at site of inoculation ▫ Local lymphadenopathy (proximal to inoculation site) ▫ Influenza-like illness with fever and myalgia
SUBACUTE TO LATE SYMPTOMS	▫ Numbness, paresthesia near exposure site, fever, conjunctivitis, abdominal pain, hepatitis, or pneumonitis ▫ CNS symptoms: headache, meningism, cranial nerve deficits, seizures, paralysis, respiratory failure, and coma
LATE-STAGE SYMPTOMS	▫ Late-stage disease is that of brainstem encephalomyelitis that may evolve into diffuse encephalomyelitis during its terminal stages.

- Incubation period: 2 days – 5 weeks (most present 5 – 21 days after exposure).
- Serology & PCR are send out tests and will not be covered under the standard CE bill.
- Refer to [ANNEX C](#) for details on the send-out test.

7.4 DISCHARGE

- Patients should be discharged with a patient information sheet.
- If discharged from CE, an early review with Paeds ID should be arranged in 24 – 48 hours.
- Discharge with [Herpes B Patient Information Sheet](#)

8. MARINE ANIMAL STINGS

8.1 TROPICAL JELLYFISH & VINEGAR

- Vinegar is the most appropriate first aid for tropical jellyfish stings (fatal, toxic stings)
- Jellyfish in tropical waters that could cause potentially fatal stings have nematocysts (stinging organelles) which inject their venom directly into capillaries, causing rapid and potentially catastrophic systemic envenoming. During the process of stinging, pieces of tentacle with unfired nematocysts may remain on the skin.
- Vinegar inhibits these unfired nematocysts so that subsequent handling does not cause harm (vinegar has no effect on venom that has already been introduced into the patient).

8.1.1 HOW?

- Flood stung area with copious amounts of vinegar for at least 30 seconds.
- Do not use fresh water, alcohol, or methylated spirits as they activate unfired nematocysts.
- Gently pick off any tentacles with a gloved hand or forceps
- If vinegar is unavailable, pick off any remnants of the tentacles and rinse stung area with sea water.

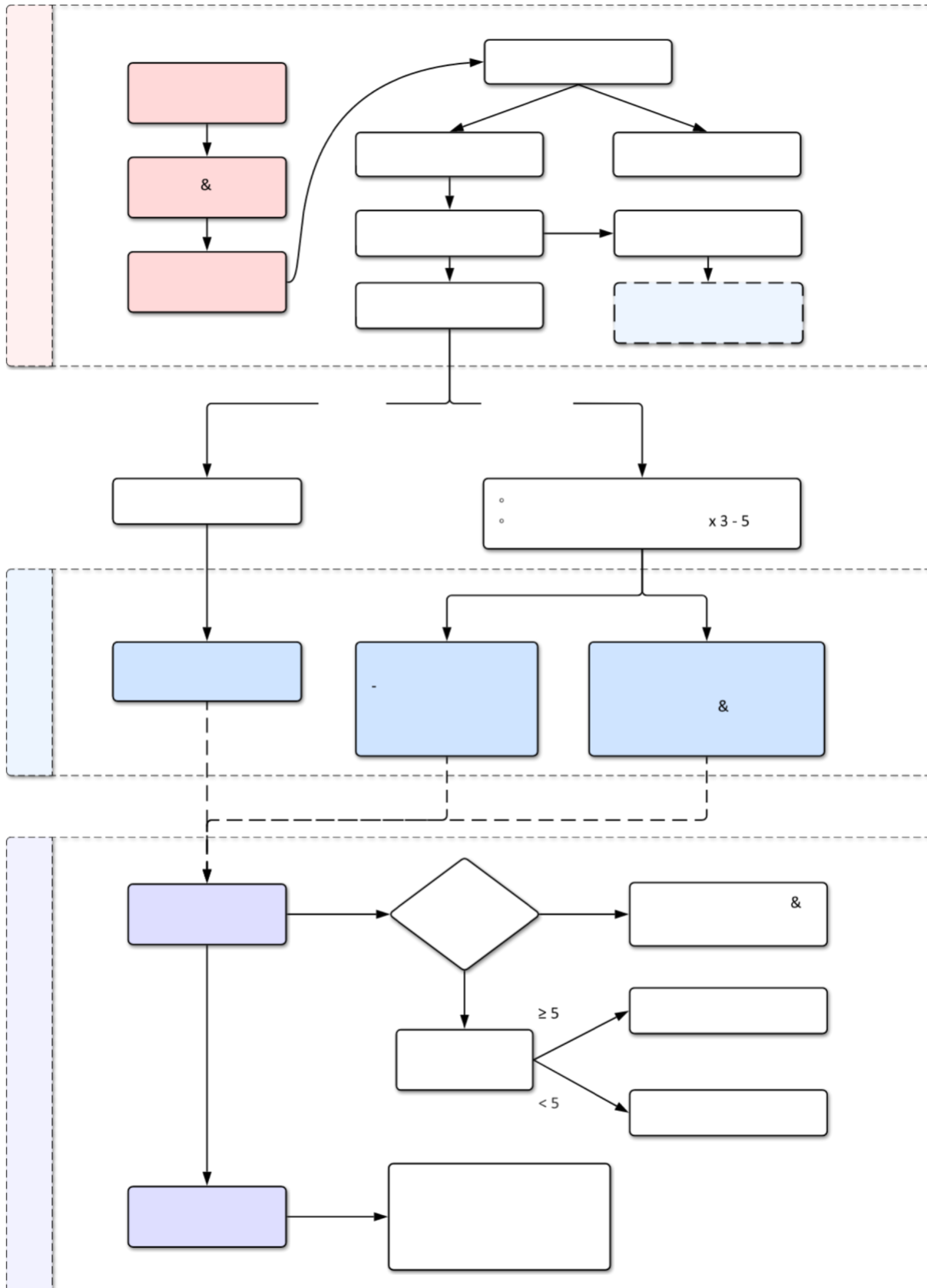
8.2 PAINFUL STINGS & WARM WATER

- When to use warm water?
 - Fish stings
 - Stingray
 - Non-tropical jellyfish
- Stings from above marine animals rarely cause severe toxic effects, mostly excruciating pain.
- Mechanism of action: Hot water is postulated to deactivate heat-labile venom toxins or modulate pain receptors.

8.2.1 HOW?

- The temperature of water should not be more than 45°C (or more than what the rescuer can tolerate).
- Place stung limb (with non-stung limb to prevent thermal scald burns) in warm water for not more than 20 minutes. Remove briefly before re-immersing.
- The cycle can be repeated for up to 2 hours if pain is still present.

ANNEX A: MANAGEMENT OF MAMMALIAN BITES



ANNEX B: TABLE OF ANTIBIOTICS FOR ANIMAL BITES

I MAMMALIAN BITES

	PASTEURELLA	ANAEROBIC
	DISCHARGE MEDICATION	
FIRST LINE ^a	PO Augmentin <u>22.5mg/kg</u> (625mg) q8H	
Penicillin allergy or Augmentin not suitable	PO Bactrim ^b <u>4mg (TMP)/kg</u> (160mg) q12H	PO clindamycin <u>10mg/kg</u> (450mg) q8H
	[†] PO ciprofloxacin ^d <u>10 – 20mg/kg</u> (500mg) q12H	PO clindamycin <u>10mg/kg</u> (450mg) q8H
	[^] PO doxycycline ^c <u>4.4mg/kg</u> (200mg) OD, D1 <u>2.2mg/kg</u> (100mg) OD, D2 – D*	PO clindamycin <u>10mg/kg</u> (450mg) q8H
	ADMITTED PATIENTS	
FIRST LINE ^a	IV Augmentin <u>30mg/kg</u> (1.2g) q8H	
Has MRSA coverage	IV ampicillin <u>50mg/kg</u> (1g) q8H	IV clindamycin <u>40mg/kg/day</u> (600mg/dose) q6H
Penicillin allergy or Augmentin not suitable	IV ceftriaxone <u>50mg/kg</u> (2g) q8H	IV clindamycin <u>40mg/kg/day</u> (600mg/dose) q6H

*Prophylaxis = 3-day course; Treatment = 5-day course.

[^]Red Book® 2018 – 2021 Report of the Committee on Infectious Diseases, 30th Ed.

[†]2021 Nelson's Pediatric Antimicrobial Therapy by AAP, 27th Edition.

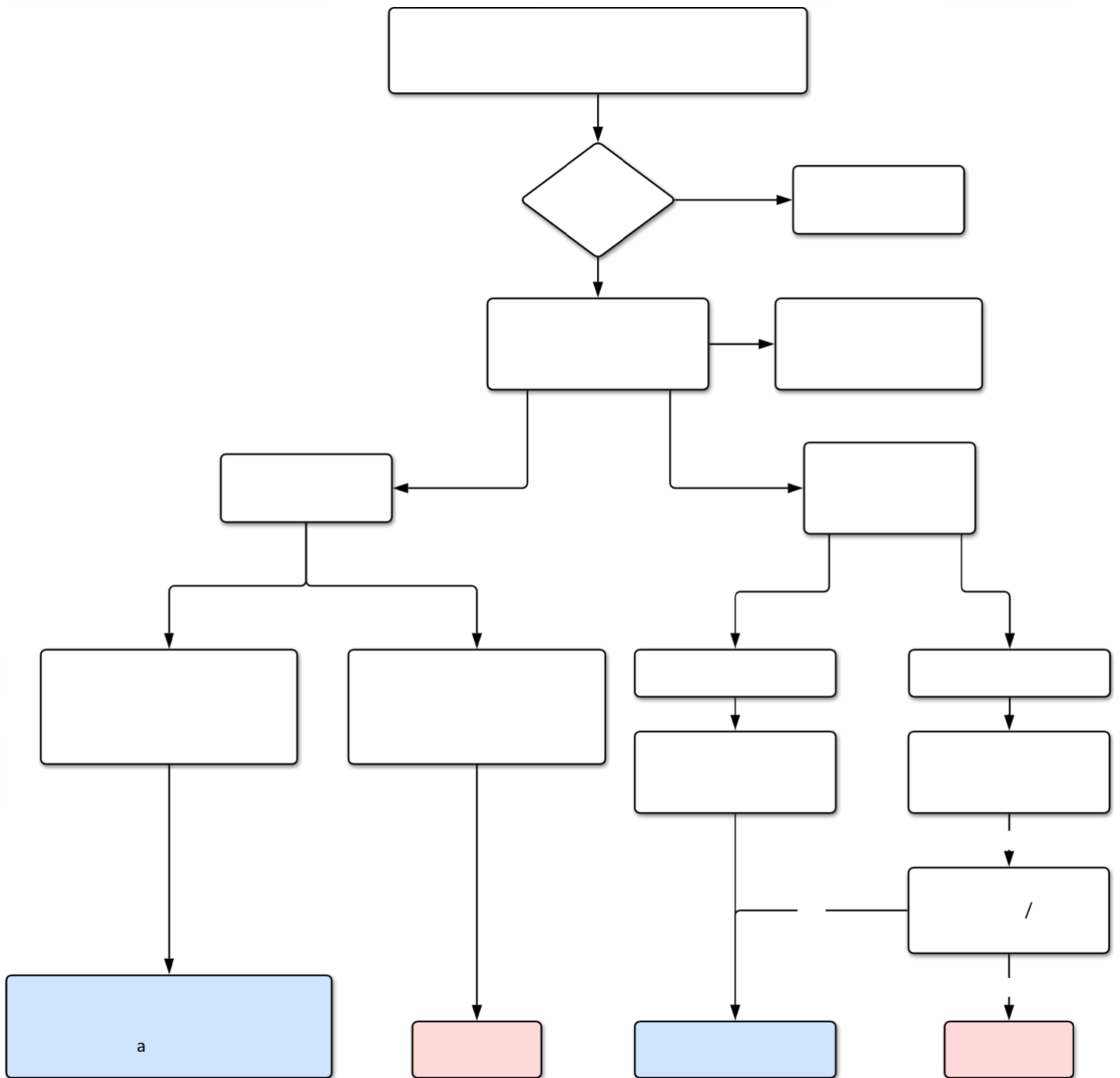
^a No MRSA cover.

^b Contraindicated in G6PD deficiency.

^c For patients > 8 years.

^d Risk of serious adverse effects (e.g., tendinopathy, CNS effects etc.).

ANNEX C: RABIES POST-EXPOSURE PROPHYLAXIS (PEP) ALGORITHM



ANNEX D: RABIES PEP AND RABIES IMMUNOGLOBULINE

I TABLE SUMMARIZING INDICATIONS FOR RABIES PEP & RIG

WHO Exposure Category	Local Tx	Immunization Status	PEP Regime	RIG
Category I Touching or feeding of animals; licks on intact skin	Required	NA.	Not required	Not required
Category II Nibbling of uncovered skin, minor scratches, or abrasions without bleeding. If bat exposure, treat as Cat III.	Required	No prior immunization received.	Day 0, 3, 7, 14	Not required
		Previously immunized. ▫ 4-dose PEP (LD [#] < 3 months) ▫ All other regimes including 4-dose PEP (LD [#] > 3 months)	Not required Day 0, 3	Not required Not required
		Immunocompromised (regardless of previous vaccination).	Day 0, 3, 7, 14, 28	Required
Category III Single or multiple transdermal bites or scratches, licks on broken skin; Contamination of mucous membrane with saliva from licks, contacts with bats. Bites to the head, neck, face, hand & genitals are considered Cat III due to extensive innervation.	Required	No prior immunization received.	Day 0, 3, 7, 14	Required
		Previously immunized: ▫ 4-dose PEP (LD [#] < 3 months) ▫ All other regimes including 4-dose PEP (LD [#] > 3 months)	Not required Day 0, 3	Not required Not required
		Immunocompromised (regardless of previous vaccination).	Day 0, 3, 7, 14, 28	Required

[#]LD – Last Dose

II RABIES VACCINE

Contraindications:

- Anaphylaxis to previous rabies vaccine.
- Severe allergy to chicken eggs.
- Allergy to neomycin, chlortetracycline, or amphotericin B.

Dose: Rabipur, Purified Chick Embryo Cell Vaccine (PCECV), 1ml.

- Injection sites: Anterolateral thigh (age < 2 years); Deltoid (age ≥ 2 years) (Do not administer into gluteal region)
- Co-administration of chloroquine is associated with a significant reduction in rabies antibody titer, though the reduced levels remain > 0.5 IU/mL (minimum acceptable rabies antibody titer level).

Adverse reactions

- Local: Pain, tenderness, or erythema.
- Systemic reactions: Headaches, fever/chills, malaise, nausea, flu-like symptoms, lymphadenopathy, or rash.

III HUMAN RABIES IMMUNOGLOBULIN (HRIG)

- HRIG is indicated in Category III exposures for immunologically naïve patients.
- HRIG should be administered once, preferably at the start of PEP and not more than 7 days following the first rabies vaccine dose as it may interfere with antibody response to the vaccine.
- Suturing should be postponed; if necessary, ensure the RIG has been applied thoroughly.
- HRIG is available in NUH: HyperRAB®, injectable solution, 300 IU per mL (*c.a.a Feb 2026*)

Contraindications

- 1st dose of rabies vaccine administered more than 7 days ago.

Instructions

- Dose = 20 IU per kg (giving more than the intended dose may interfere with antibody response to rabies vaccination).
- If anatomically feasible, the full dose (or at least more than half the dose) of HRIG should be infiltrated into the depth of the wound & around the wound.
- For wounds that are large, or multiple wounds, the maximum calculated volume of HRIG may be further diluted in a compatible diluent (see drug insert) to infiltrate all wounds effectively.
- Inject the remainder, if any, intramuscularly (into the deltoid muscle or lateral thigh muscle) at an anatomical site distant from the site of vaccine administration.
- Use separate syringes, needles, and anatomical injection sites for HRIG and for rabies vaccine.

Adverse Reactions/Side Effects

- Local reactions: Pain & tenderness, hypoesthesia, erythema, and induration.
- Systemic reactions: Headaches, fever, malaise, nausea, and weakness.
- Risk of blood borne infections.

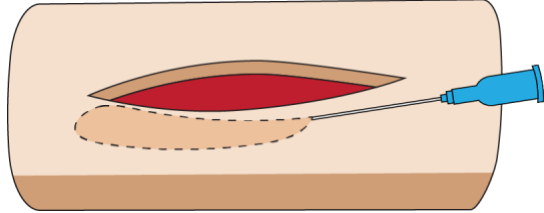
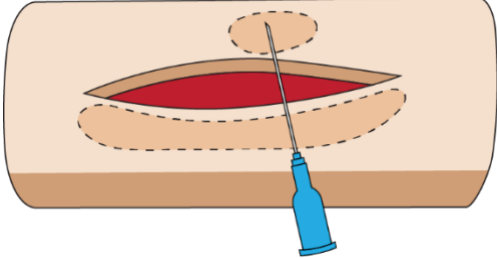
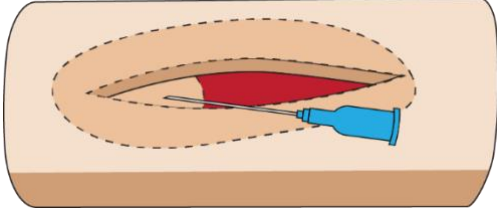
Post-Administration Monitoring/Advice

- Observe patient for 60 minutes after administering HRIG.
- Monitor HR, RR, BP and SpO₂ at 0, 30, and 60 minutes.
- Discharge with memo on deferring live vaccinations. Live vaccines (MMR, varicella, rotavirus) should be deferred till 4 months after HRIG administration.

IV ADMINISTERING RABIES IMMUNOGLOBULIN (HRIG)

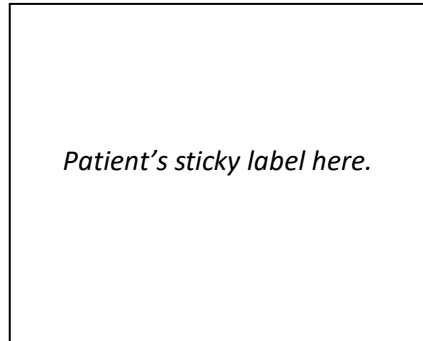
1. Verify the patient's weight and previous immunization records.
2. Check that the patient has not previously received pre-exposure prophylaxis.
3. If part of post-exposure prophylaxis (PEP), ensure that it is within 7 days of the 1st dose of rabies vaccine.
4. Dose of HRIG = 20 IU/kg
5. Infiltrate the entire volume into the depth of the wound & around the wound if anatomically feasible (similar to infiltrating local anesthesia to a wound)
6. For large or multiple wounds, dilute HRIG in an appropriate diluent (see drug insert) to a volume sufficient to infiltrate all wounds effectively.
7. Separate needles, syringes & anatomical injection sites for RIG & the rabies vaccine.

How to Infiltrate HRIG?

STEPS	DIAGRAM
1. Pick an injection site approximately at the mid-point of the wound. 2. Advance the needle along the edge of the wound and infiltrate RIG as the needle is withdrawn (distal to proximal). Without completely withdrawing the needle, repeat the same to the rest of the ipsilateral wound edge.	
3. Without withdrawing the needle completely, advance the needle tip to the contralateral wound edge and infiltrate around the contralateral wound edge. 4. While infiltrating around the wound, HRIG should spill into the base of the wound.	
5. Infiltrate into the base of the wound.	

MEMORANDUM

Dear colleague,



The above-mentioned patient had a WHO category III rabies exposure and received the rabies immunoglobulin (RIG) as part of rabies post-exposure prophylaxis. RIG may interfere with antibody response to live vaccines (e.g., MMR, varicella, rotavirus). All live vaccines should be deferred to 4 months from when RIG was administered.

No live vaccines from: (start date) _____ to
(end date) _____ (inclusive).

Best regards,

Name: _____

MCR: _____

Designation: _____

ANNEX E: HERPES B VIRUS/MONKEY BITES

I HERPES B LABORATORY PROCEDURES

TEST	DETAILS
Herpes B Simian, Serum	Container: Red Top Volume: 6ml (minimum 1ml) Send with National B Virus Request Form (available on intranet)
Herpes B Simian, Swab	Tube containing viral transport medium (VTM) 1. Swab each collection site with a separate sterile swab 2. Each swab to be placed into separate sample tubes with viral transport media

Samples are sent out to the National B Virus Resource Center, Viral Immunology Center in Georgia State University for Human Herpes B Simian testing.

HERPES B PATIENT INFORMATION SHEET

Dear Sir/ Mdm,

You may have been exposed to B virus infection. This condition is potentially dangerous. Please seek medical treatment immediately if you develop the following symptoms.

- At the site of your injury, please look out for the following:
 - Vesicular (fluid containing sac) rash
 - Tingling
 - Itching
 - Pain
 - Numbness at the site or proximal to it, along the affected extremity
 - Swollen lymph nodes proximal to the site of inoculation
- Generalized symptoms:
 - Flu-like illness with fever, chills, muscle aches, malaise
 - Nausea and/or Vomiting
 - Neck stiffness
 - Persistent headache
 - Confusion
 - Double vision
 - Difficulty swallowing
 - Dizziness

Please show this document to your family members. If you have any queries, please call 6779 5999 to speak with the Emergency Physician on duty.

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